

AP CSP | Lesson 25: Fault Tolerance

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 45 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Q5 requires exactly TWO choices. Show paths, calculations and reasons. Treat all networks and timings as the stated classroom models.

1. What is redundancy in a system? [1 mark]

- A. A promise that nothing fails
- B. Extra components that can mitigate other failures
- C. The removal of all alternative paths
- D. A larger font on a network map

Answer: _____

2. Which observation demonstrates fault tolerance for S-to-T communication? [1 mark]

- A. No component has ever failed.
- B. A diagram has many colors.
- C. Communication continues after a specified link fails.
- D. The receiver's software never needs testing.

Answer: _____

3. S-A-C-T loses A-C. Working edges S-A, A-D, D-T remain. Which path works? [1 mark]

- A. S-A-D-T
- B. S-A-C-T
- C. S-T, even without that edge
- D. S-C-T, even without S-C

Answer: _____

4. S has only two incident links, S-A and S-B. Both fail. What follows? [1 mark]

- A. Remote extra links automatically reconnect S.
- B. S can jump to T.
- C. Only bandwidth changes.
- D. S cannot reach any other node.

Answer: _____

5. Which TWO statements are correct? [1 mark]

- A. Redundancy may require more equipment or maintenance.
- B. Any two drawn routes fail independently.
- C. A path guarantees a functioning application.
- D. Multiple routes can share a physical failure risk.

Answer: _____

6. Route X has 3 hops and route Y has 4. What can be concluded about real delay from this alone? [1 mark]

- A. Y is always slower by 1 second.
- B. Actual delay is not determined by hop count alone.
- C. Both have the same bandwidth.
- D. X can never lose packets.

Answer: _____

AP CSP | Lesson 25: Fault Tolerance

Written responses | Questions 7-9

7. [4 marks]

All edges are undirected: S-A, S-B, A-C, A-D, B-D, C-T, D-T, C-D. (a) Give two different S-to-T paths, including one avoiding A-C. (b) Give each hop count. (c) Explain how those alternatives help if A-C fails.

8. [4 marks]

Use the graph in Q7. Only C-D fails. (a) Give a working route. (b) Count its hops. (c) Does S-A-C-T remain valid? Explain. (d) Does unchanged routing mean the failed edge did not exist?

9. [4 marks]

Use Q7's graph. Disable C-T and D-T together. (a) Is S-to-T communication possible? (b) Explain structurally. (c) Give a different two-edge set that also disconnects S and T. (d) Explain why extra edges elsewhere are insufficient.

AP CSP | Lesson 25: Fault Tolerance

Written responses | Questions 10-12

10. [4 marks]

Use Q7's graph. Disable S-B, A-D and C-T. (a) Give the surviving S-to-T route. (b) Count hops. (c) At exactly 1 time unit per hop, give its cost. (d) Why is this not a measured Internet delay?

11. [4 marks]

Two routes use cables in the same tunnel. (a) Identify a shared risk. (b) Explain why two drawn routes do not ensure independent failure. (c) Propose a relevant improvement. (d) Give one associated resource cost.

12. [4 marks]

A team observes a route after one link failure and says the whole service can never fail. (a) State what their evidence supports. (b) Give a network counterexample using Q7. (c) Give a non-path limitation. (d) Rewrite the claim accurately.
